AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2016

General Mathematics Paper II

Time: 2 hours 20 minutes Marks: 45

AKU EXAMINATIONS 2016

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL questions. Choices are specified inside the paper

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use colour pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

In a class of 140 students, 70% of the students wear glasses while 5% wear contact lenses. Find the number of students that wear

i. glasses. (1 Mark)

ii. contact lenses. (1 Mark)

iii. neither glasses nor lenses. (1 Mark)

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(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 4 Marks)

a. The total amount of Zakat paid by two brothers Ahmar and Bilal is Rs 50,000. Ahmar paid Zakat of Rs 23,000 on his yearly savings. Find

i. the amount of Zakat paid by Bilal. (1 Mark)

ii. the individual yearly savings of Ahmar and Bilal. (2 Marks)

iii. the ratio of Zakat paid by Ahmar and Bilal. (1 Mark)

b. Rehman possesses a property worth Rs 500,000 and wishes to distribute his property among his legal heirs that include 1 son and 3 daughters. Find the share of each legal heir.

(Note: The son gets twice as much as a daughter gets.)

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(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 5 Marks)

a. Mr Ali sold his land for Rs 9,000,000. His loss amounted to two fifths of his cost. Find

i. the amount the land cost him. (3 Marks)

ii. the percentage loss. (2 Marks)

b. A fruit vendor bought 600 apples for Rs 4,800. He spent Rs 500 on transportation. Find the price at which he should sell each apple to get a total profit of Rs 1,900. Also, find his profit percentage. (5 Marks)

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Q.4. (Total 3 Marks)

Fahad leased a car worth Rs 500,000 from a leasing company with a down payment of Rs 200,000. If the rate of simple interest is 7.5% per annum for a period of 2 years, find the interest Fahad will have to pay on the remaining amount.

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Q.5. (Total 5 Marks)

Saad earns a basic monthly salary of Rs 20,000. As a salesman, he must sell Rs 300,000 worth of items per month but if he sells beyond this monthly quota, he makes a 4% commission on the additional sale. Saad sells Rs 460,000 worth of items in a certain month.

i. Find his total (gross) salary for this month. (3 Marks)

ii. Saad needs to pay income tax on his basic salary. If first Rs 15,000 is tax exempted, find the amount of tax that he will pay on his basic salary if the rate of income tax is 20%. (2 Marks)

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(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a. Express $\frac{\left(2x^7\right)^3}{\sqrt{4x^8}}$ in its simplest form. (5 Marks)

b. Simplify the following and give your answer as an integer.

i. $\log_5 25$ (3 Marks)

ii. $\log_5 \frac{5}{2} + \log_5 2$ (2 Marks)

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Q.7. (Total 4 Marks)

In a geometric sequence, the first term is 2 and the fourth term is 250. Find the

i. common ratio of this sequence. (3 Marks)

ii. 7th term of this sequence. (1 Mark)

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(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 5 Marks)

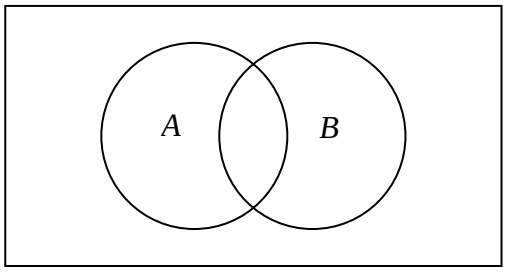
a. For sets $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{2, 4, 6, 8, 10\}$, verify that $(A \cap B) - A = (A \cap B) - B$. Also find $(A \cup B) - A$.

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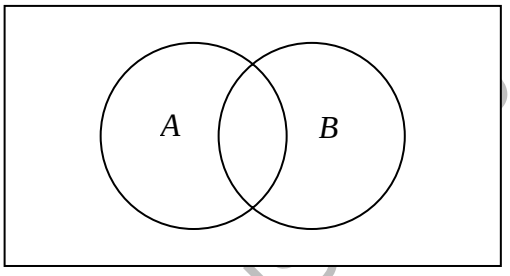
(ATTEMPT EITHER PART a OR PART b OF Q.8.)

b. If A and B are any two non-empty and mutually overlapping sets, then represent the following by shading the given Venn diagrams.

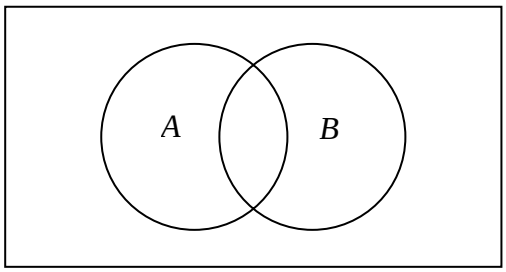
i. $A \cup B$



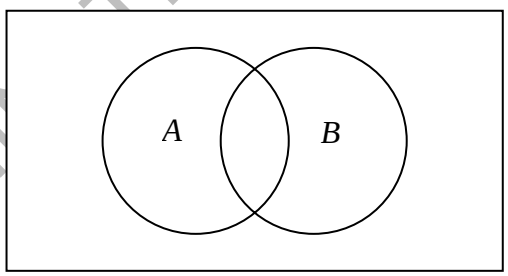
ii. $A \cap B$



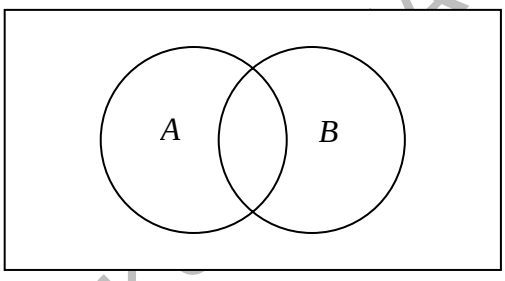
iii. $B - A$



iv. $A - B$



v. A^c



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Q.9. (Total 5 Marks)

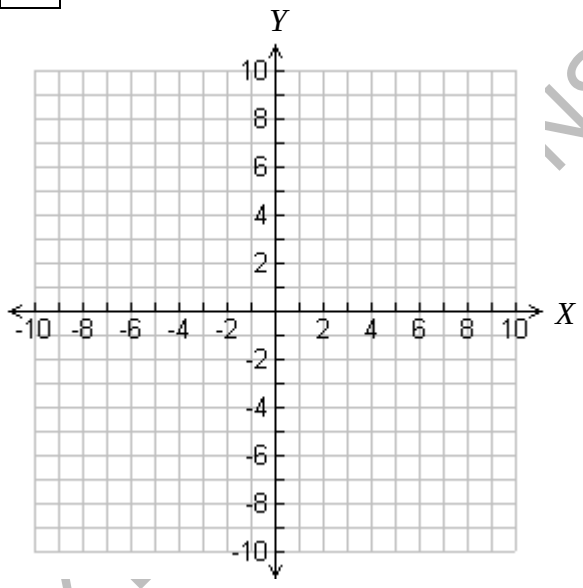
On the given graph, draw the following lines

i. $x = 6$ (1 Mark)

ii. $y = -2$ (1 Mark)

iii. $y = x + 1$ (2 Marks)

x	1	2	3
$y = x + 1$	2	3	4



iv. Name the figure obtained by drawing the lines (i), (ii) and (iii) on the graph.

(1 Mark)

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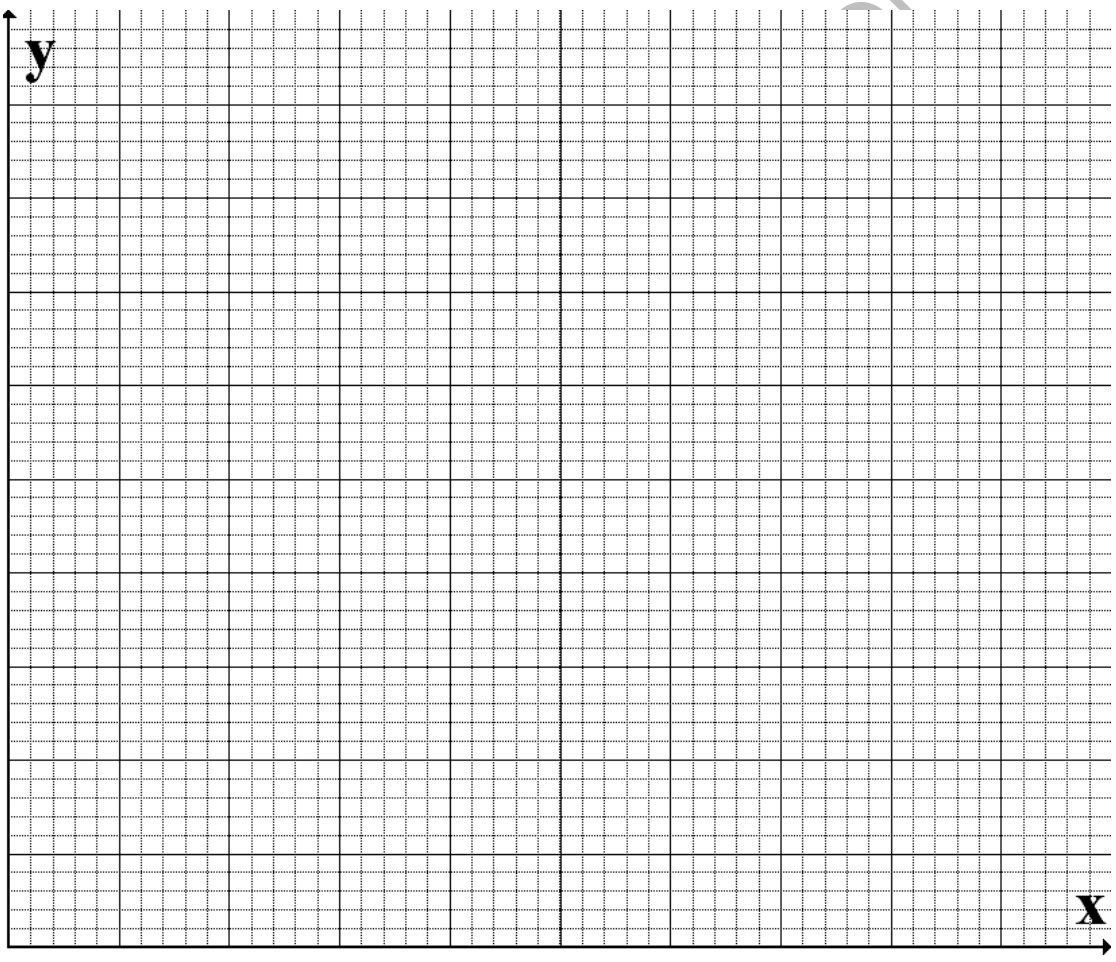
(ATTEMPT EITHER PART a OR PART b OF Q.10.)

Q.10. (Total 6 Marks)

a. The given data shows the weight of 25 items in kilograms.

Weight (kg)	0 – 2	2 – 4	4 – 6	6 – 8
Frequency	6	8	7	4

On the grid given below, represent this data in the form of a histogram and find the mode graphically.



b. The following data shows the number of shirts each of Sara's 20 friends own.

9, 15, 11, 12, 3, 5, 10, 20, 14, 6, 8, 8, 12, 12, 18, 15, 6, 9, 18, 11

The frequency distribution table for this data is shown below. Complete this table and find the mean number of shirts owned by Sara's 20 friends.

Number of Shirts	**Tally Marks**	**Frequency**	**Class Mark**	
1 – 5				
6 – 10				
11 – 15				
16 – 20				
Total	–		–	

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